

MA3236 NONLINEAR PROGRAMMING

Semester 1, AY 2026/2027

Tutorial 5

Review: Lecture 10 — Steepest descent

Let $f : \mathbf{R}^n \rightarrow \mathbf{R}$, $\mathbf{g}^{(k)} = \nabla f(\mathbf{x}^{(k)})$, and $\mathbf{d}^{(k)} = -\mathbf{g}^{(k)}$. For $\mathbf{g}^{(k)} \neq \mathbf{0}$, the direction is descending and a line-search step is

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}, \quad \alpha^{(k)} > 0.$$

- **Smooth quadratic upper model.** If ∇f is L -Lipschitz, then

$$f(\mathbf{y}) \leq M_L(\mathbf{y}; \mathbf{x}) := f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) + \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|^2.$$

The model minimizer is $\mathbf{x} - L^{-1} \nabla f(\mathbf{x})$, and

$$f\left(\mathbf{x} - \frac{1}{L} \nabla f(\mathbf{x})\right) \leq f(\mathbf{x}) - \frac{1}{2L} \|\nabla f(\mathbf{x})\|^2.$$

Thus $\alpha = 1/L$ satisfies the sufficient-decrease test in Armijo backtracking line search with $\sigma = 1/2$; this step bound is sufficient, not necessary.

- **Exact search on a positive-definite quadratic.** For $q(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} - \mathbf{c}^T \mathbf{x}$, with symmetric $\mathbf{Q} \succ 0$, the SD direction is $\mathbf{d} = -\mathbf{g}$ and

$$\alpha_{\text{exact}} = \frac{\|\mathbf{g}\|^2}{\mathbf{g}^T \mathbf{Q} \mathbf{g}}.$$

Successive gradients are orthogonal in exact arithmetic.

- **Rate and preconditioning.** If f is μ -strongly convex and L -smooth, the $1/L$ -step SD method satisfies

$$E^{(k+1)} \leq \left(1 - \frac{1}{\kappa}\right) E^{(k)},$$

where $E^{(k)} = f(\mathbf{x}^{(k)}) - f^*$ and $\kappa = L/\mu$. For a quadratic, an exact-search SD worst-case gap factor is $((\kappa - 1)/(\kappa + 1))^2$. A symmetric PD preconditioner $\mathbf{B}$ uses $\mathbf{d} = -\mathbf{B}^{-1} \mathbf{g}$; its effective condition number is that of $\mathbf{B}^{-1/2} \mathbf{Q} \mathbf{B}^{-1/2}$.

Review: Lecture 11 — Conjugate directions and CG

For the same quadratic, write $\mathbf{g}^{(k)} = \mathbf{Q}\mathbf{x}^{(k)} - \mathbf{c}$. Directions $\mathbf{d}^{(i)}$ and $\mathbf{d}^{(j)}$ are $\mathbf{Q}$ -conjugate when $(\mathbf{d}^{(i)})^T \mathbf{Q} \mathbf{d}^{(j)} = 0$ for $i \neq j$. Exact line searches along conjugate directions preserve all earlier line-search optimality conditions:

$$\alpha^{(k)} = -\frac{(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)}}{(\mathbf{d}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k)}}.$$

After $k+1$ independent conjugate searches, the iterate minimizes q over the affine subspace

$$\mathbf{x}^{(0)} + \text{span}\{\mathbf{d}^{(0)}, \dots, \mathbf{d}^{(k)}\}.$$

- **Conjugate gradient (CG).** It constructs the conjugate directions recursively:

$$\begin{aligned} \mathbf{d}^{(0)} &= -\mathbf{g}^{(0)}, & \alpha^{(k)} &= \frac{\|\mathbf{g}^{(k)}\|^2}{(\mathbf{d}^{(k)})^T \mathbf{Q} \mathbf{d}^{(k)}}, \\ \mathbf{x}^{(k+1)} &= \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}, & \mathbf{g}^{(k+1)} &= \mathbf{g}^{(k)} + \alpha^{(k)} \mathbf{Q} \mathbf{d}^{(k)}, \\ \beta^{(k+1)} &= \frac{\|\mathbf{g}^{(k+1)}\|^2}{\|\mathbf{g}^{(k)}\|^2}, & \mathbf{d}^{(k+1)} &= -\mathbf{g}^{(k+1)} + \beta^{(k+1)} \mathbf{d}^{(k)}. \end{aligned}$$

In exact arithmetic, CG terminates in at most n updates; fewer can suffice when the initial error involves fewer distinct eigenvalues. Finite precision can spoil exact conjugacy and termination.

- **Preconditioned CG (PCG).** For a fixed symmetric PD $\mathbf{B}$, set $\mathbf{z}^{(k)} = \mathbf{B}^{-1} \mathbf{g}^{(k)}$ and start with $\mathbf{d}^{(0)} = -\mathbf{z}^{(0)}$. Its first step is

$$\alpha^{(0)} = \frac{(\mathbf{g}^{(0)})^T \mathbf{z}^{(0)}}{(\mathbf{d}^{(0)})^T \mathbf{Q} \mathbf{d}^{(0)}}.$$

PCG is CG applied in transformed coordinates to $\mathbf{B}^{-1/2} \mathbf{Q} \mathbf{B}^{-1/2}$. A better effective condition number can reduce iterations, but applying $\mathbf{B}^{-1}$ also has a cost.

1. **A smooth upper model and one gradient step.** Let

$$q(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x}, \quad \mathbf{Q} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}, \quad \mathbf{x}^{(0)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

- (a) Find the eigenvalues of $\mathbf{Q}$, and hence valid sharp L -smoothness and strong-convexity constants. At $\mathbf{x}^{(0)}$, write the quadratic upper model $M_L(\mathbf{y}; \mathbf{x}^{(0)})$, prove that it majorizes q , and find its minimizer.
- (b) Compute the resulting SD iterate $\mathbf{x}^{(1)}$ and both function values.
- (c) Check the Armijo condition with $\sigma = 1/2$ and $\alpha = 1/L$. Compare the actual gap ratio with the $1 - \mu/L$ bound.

Solution. The eigenvalues of $\mathbf{Q}$ are 2 and 4, so $\mu = 2$, $L = 4$, and $\nabla q(\mathbf{x}^{(0)}) = (3, 1)^T$. Put $\mathbf{s} = \mathbf{y} - \mathbf{x}^{(0)}$. The model is

$$M_4(\mathbf{y}; \mathbf{x}^{(0)}) = \frac{3}{2} + \begin{pmatrix} 3 & 1 \end{pmatrix} \mathbf{s} + 2\|\mathbf{s}\|^2.$$

Because

$$M_4(\mathbf{y}; \mathbf{x}^{(0)}) - q(\mathbf{y}) = \frac{1}{2}\mathbf{s}^T (4I - \mathbf{Q}) \mathbf{s} = \frac{1}{2}(s_1 - s_2)^2 \geq 0,$$

it majorizes q , with equality at $\mathbf{y} = \mathbf{x}^{(0)}$. Its unique minimizer is

$$\mathbf{x}^{(1)} = \mathbf{x}^{(0)} - \frac{1}{4}\nabla q(\mathbf{x}^{(0)}) = \begin{pmatrix} 1/4 \\ -1/4 \end{pmatrix}.$$

Direct calculation gives

$$q(\mathbf{x}^{(0)}) = \frac{3}{2}, \quad q(\mathbf{x}^{(1)}) = \frac{1}{8}, \quad q(\mathbf{x}^{(0)}) - \frac{\|\nabla q(\mathbf{x}^{(0)})\|^2}{8} = \frac{1}{4}.$$

Thus $1/8 \leq 1/4$. For $\mathbf{d}^{(0)} = -\nabla q(\mathbf{x}^{(0)})$, the Armijo right-hand side is $q(\mathbf{x}^{(0)}) + \frac{1}{2} \frac{1}{4} (\mathbf{g}^{(0)})^T \mathbf{d}^{(0)} = 1/4$; the step is accepted. Since $q^* = 0$, the actual gap ratio is $1/12$, smaller than $1 - \mu/L = 1/2$.

2. **Exact-search SD, a rate, and preconditioning.** Consider

$$q(\mathbf{x}) = \frac{1}{2}(x_1^2 + 4x_2^2), \quad \mathbf{x}^{(0)} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}.$$

- (a) Compute $\alpha^{(0)}, \mathbf{x}^{(1)}, \alpha^{(1)}$, and $\mathbf{x}^{(2)}$ under exact-line-search SD. Verify that $(\mathbf{g}^{(0)})^T \mathbf{g}^{(1)} = 0$.
- (b) Prove by induction that

$$\alpha^{(k)} = \frac{2}{5}, \quad \mathbf{x}^{(k)} = \left(\frac{3}{5}\right)^k \begin{pmatrix} 4 \\ (-1)^k \end{pmatrix}.$$

Also show that

$$\frac{q(\mathbf{x}^{(k)})}{q(\mathbf{x}^{(0)})} = \left(\frac{9}{25}\right)^k.$$

Relate the factor to the condition number of $\mathbf{Q}$.

- (c) Find the first k for which $q(\mathbf{x}^{(k)})/q(\mathbf{x}^{(0)}) \leq 10^{-3}$. What happens if preconditioned SD uses $\mathbf{B} = \text{diag}(1, 4)$ with exact line search?

Solution. Here $\mathbf{Q} = \text{diag}(1, 4)$ and $\mathbf{g}^{(0)} = (4, 4)^T$. The exact step is

$$\alpha^{(0)} = \frac{\|\mathbf{g}^{(0)}\|^2}{(\mathbf{g}^{(0)})^T \mathbf{Q} \mathbf{g}^{(0)}} = \frac{32}{80} = \frac{2}{5}.$$

Thus

$$\mathbf{x}^{(1)} = \begin{pmatrix} 12/5 \\ -3/5 \end{pmatrix}, \quad \mathbf{g}^{(1)} = \begin{pmatrix} 12/5 \\ -12/5 \end{pmatrix}, \quad (\mathbf{g}^{(0)})^T \mathbf{g}^{(1)} = 0.$$

The two components of $\mathbf{g}^{(1)}$ again have equal absolute value, so $\alpha^{(1)} = 2/5$ and $\mathbf{x}^{(2)} = (36/25, 9/25)^T$. More generally, the displayed formula for $\mathbf{x}^{(k)}$ gives

$$\mathbf{g}^{(k)} = 4 \left(\frac{3}{5}\right)^k \begin{pmatrix} 1 \\ (-1)^k \end{pmatrix}.$$

The exact-step formula therefore always gives $2/5$, and

$$\mathbf{x}^{(k)} - \frac{2}{5} \mathbf{g}^{(k)} = \left(\frac{3}{5}\right)^{k+1} \begin{pmatrix} 4 \\ (-1)^{k+1} \end{pmatrix}.$$

This proves the induction. Since $q(\mathbf{x}^{(0)}) = 10$,

$$q(\mathbf{x}^{(k)}) = 10 \left(\frac{9}{25}\right)^k.$$

The condition number is 4, and $((4-1)/(4+1))^2 = 9/25$: this example attains the exact-search SD worst-case factor. As $(9/25)^6 > 10^{-3}$ but $(9/25)^7 < 10^{-3}$, the first acceptable index is $k = 7$. For $\mathbf{B} = \mathbf{Q}$, the preconditioned direction is $-\mathbf{B}^{-1} \mathbf{g} = -\mathbf{x}$, so exact search takes $\alpha = 1$ and reaches $\mathbf{0}$ in one update.

3. **Constructing two conjugate directions.** Let

$$q(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x} - \mathbf{c}^T \mathbf{x}, \quad \mathbf{Q} = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Start from

$$\mathbf{x}^{(0)} = \mathbf{0}, \quad \mathbf{d}^{(0)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

- (a) Verify that $\mathbf{Q} \succ 0$. Find the unique direction of the form $\mathbf{d}^{(1)} = (t, 1)^T$ that is $\mathbf{Q}$ -conjugate to $\mathbf{d}^{(0)}$.
- (b) Perform exact line searches along $\mathbf{d}^{(0)}$ and then $\mathbf{d}^{(1)}$. Give both step lengths, both iterates, and the final gradient.
- (c) Explain using gradient orthogonality why the second step does not undo the first line minimization, and identify the affine subspace minimized over after both steps.

Solution. The leading principal minors are 4 and 7, so $\mathbf{Q} \succ 0$. Conjugacy requires

$$(\mathbf{d}^{(0)})^T \mathbf{Q} \mathbf{d}^{(1)} = 4t + 1 = 0, \quad \mathbf{d}^{(1)} = \begin{pmatrix} -1/4 \\ 1 \end{pmatrix}.$$

Initially $\mathbf{g}^{(0)} = -\mathbf{c} = (-1, -2)^T$, hence

$$\alpha^{(0)} = -\frac{(\mathbf{g}^{(0)})^T \mathbf{d}^{(0)}}{(\mathbf{d}^{(0)})^T \mathbf{Q} \mathbf{d}^{(0)}} = \frac{1}{4}, \quad \mathbf{x}^{(1)} = \begin{pmatrix} 1/4 \\ 0 \end{pmatrix}, \quad \mathbf{g}^{(1)} = \begin{pmatrix} 0 \\ -7/4 \end{pmatrix}.$$

As $\mathbf{Q} \mathbf{d}^{(1)} = (0, 7/4)^T$,

$$\alpha^{(1)} = -\frac{(\mathbf{g}^{(1)})^T \mathbf{d}^{(1)}}{(\mathbf{d}^{(1)})^T \mathbf{Q} \mathbf{d}^{(1)}} = 1, \quad \mathbf{x}^{(2)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathbf{g}^{(2)} = \mathbf{0}.$$

The first search gives $(\mathbf{g}^{(1)})^T \mathbf{d}^{(0)} = 0$; after the second,

$$(\mathbf{g}^{(2)})^T \mathbf{d}^{(0)} = (\mathbf{g}^{(1)})^T \mathbf{d}^{(0)} + \alpha^{(1)} (\mathbf{d}^{(1)})^T \mathbf{Q} \mathbf{d}^{(0)} = 0.$$

The second search also gives $(\mathbf{g}^{(2)})^T \mathbf{d}^{(1)} = 0$. The two independent directions span $\mathbf{R}^2$, so $\mathbf{x}^{(2)}$ minimizes q over $\mathbf{x}^{(0)} + \text{span}\{\mathbf{d}^{(0)}, \mathbf{d}^{(1)}\} = \mathbf{R}^2$.

4. **CG versus preconditioned CG.** For $\frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} - \mathbf{c}^T\mathbf{x}$, use

$$\mathbf{Q} = \text{diag}(1, 4, 9), \quad \mathbf{c} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{x}^{(0)} = \mathbf{0}, \quad \mathbf{B} = \text{diag}(1, 4, 3).$$

- (a) Find $\kappa(\mathbf{Q})$ and the effective condition number $\kappa(\mathbf{B}^{-1/2}\mathbf{Q}\mathbf{B}^{-1/2})$.
- (b) Compute the first direction, exact step length, and iterate for ordinary CG and for PCG with $\mathbf{B}$.
- (c) In exact arithmetic, how many updates does each method need to reach the minimizer? Justify your answer from the active eigenvalues, and state one reason the conclusion can fail in floating-point arithmetic.

Solution. The eigenvalues of $\mathbf{Q}$ are 1, 4, 9, so $\kappa(\mathbf{Q}) = 9$. The transformed matrix is

$$\mathbf{B}^{-1/2}\mathbf{Q}\mathbf{B}^{-1/2} = \text{diag}(1, 1, 3),$$

with effective condition number 3. At $\mathbf{x}^{(0)} = \mathbf{0}$, $\mathbf{g}^{(0)} = -\mathbf{c}$. Ordinary CG starts with $\mathbf{d}_{\text{CG}}^{(0)} = (1, 1, 1)^T$ and

$$\alpha_{\text{CG}}^{(0)} = \frac{\|\mathbf{g}^{(0)}\|^2}{(\mathbf{d}_{\text{CG}}^{(0)})^T\mathbf{Q}\mathbf{d}_{\text{CG}}^{(0)}} = \frac{3}{14}, \quad \mathbf{x}_{\text{CG}}^{(1)} = \frac{3}{14} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

For PCG,

$$\mathbf{z}^{(0)} = \mathbf{B}^{-1}\mathbf{g}^{(0)} = -\begin{pmatrix} 1 \\ 1/4 \\ 1/3 \end{pmatrix}, \quad \mathbf{d}_{\text{PCG}}^{(0)} = -\mathbf{z}^{(0)} = \begin{pmatrix} 1 \\ 1/4 \\ 1/3 \end{pmatrix}.$$

Consequently,

$$(\mathbf{g}^{(0)})^T\mathbf{z}^{(0)} = \frac{19}{12}, \quad (\mathbf{d}_{\text{PCG}}^{(0)})^T\mathbf{Q}\mathbf{d}_{\text{PCG}}^{(0)} = \frac{9}{4}, \quad \alpha_{\text{PCG}}^{(0)} = \frac{19}{27}, \quad \mathbf{x}_{\text{PCG}}^{(1)} = \begin{pmatrix} 19/27 \\ 19/108 \\ 19/81 \end{pmatrix}.$$

The ordinary system has three distinct eigenvalues and the initial error has a nonzero component in each eigendirection. Therefore CG needs exactly three updates in exact arithmetic. The transformed system has only two distinct active eigenvalues, 1 and 3, so PCG needs exactly two. Floating-point errors can destroy conjugacy and hence exact finite termination.